DSAI541 Assignment 06

April 15, 2026

0.1 Assignment 6 - Deadline: April 15, 2026, Wed 11pm

DSAI 541 Spring 2026 Complete the assignment below and upload both the .ipynb file and its pdf to https://moodle.bogazici.edu.tr by the deadline given above. The submission page on Moodle will close automatically after this date and time and no late homeworks will be accepted.

To make a pdf, this may work if you're using Jupyter notebook on browser: Hit CMD+P or CTRL+P, and save it as PDF. You may also use other options from the File menu. (You can use Google Colab, but at the end you should download the ipynb notebook to submit it on Moodle.)

[2]:
```
# Run this cell first

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from IPython.display import HTML
from sklearn.model_selection import train_test_split

# Set the display option to show all rows scrolling with a slider
pd.set_option('display.max_rows', None)
# To disable this, run the line below:
# pd.reset_option('display.max_rows')
```

1 Problem 1: LSTM with Jena weather dataset (60 pts)

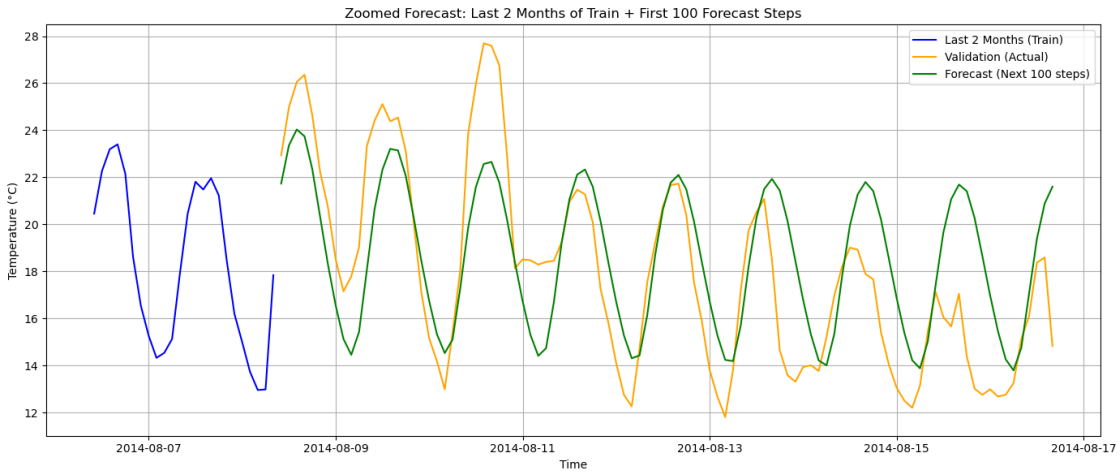
(You can use Tensorflow or PyTorch in this assignment.)

We will forecast temperature using past temperature data, where the data is resampled hourly (unlike in class, where we used two-hour intervals).

Part a) Load the Jena weather dataset. Do data cleaning (filling missing values, interpolation etc) just as we did in class (Lecture 10). Then, generate `data_1H`, where the data points are spaced 1 hour apart (in class, we had generated `data_2H`). The sequence length should cover past 6 days, so modify `sequence_length` accordingly.

Part b) Train an LSTM model as we did in class by using only the "T (degC)" column to predict the next temperature value. Make sure you use early stopping. Plot your train and validation curves. Also report MSE, MAE and MAPE on the test set.

Part c) Use the data from the latest part from the train set to predict next 100 data points via recursive forecasting as we did in class. Make a plot similar to the below. (Here using the code from Lecture 10 notebook may be easier than using ChatGPT.)



Part d) Repeat all steps, but use a slightly different model (different amount of LSTM neurons, layers and/or dropout) and make the plot again as in Part c).

Import of necessery libraries and other stuffs

[3]:
```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset
# Set random seeds for reproducibility
torch.manual_seed(42)
np.random.seed(42)
# Check if GPU is available
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {device}')
```

```
Using device: cuda
```

[4]:
```
from sklearn.preprocessing import StandardScaler
import copy
```

Part A)

[5]:
```
data = pd.read_csv('jena_climate_2009_2016.csv')
data['Date Time'] = pd.to_datetime(data['Date Time'], format='%d.%m.%Y %H:%M:
 ↪%S')


data.head()
```

[5]:
```
            Date Time  p (mbar)  T (degC)  Tpot (K)  Tdew (degC)  rh (%)  \
0 2009-01-01 00:10:00    996.52     -8.02    265.40        -8.90    93.3
1 2009-01-01 00:20:00    996.57     -8.41    265.01        -9.28    93.4
2 2009-01-01 00:30:00    996.53     -8.51    264.91        -9.31    93.9
3 2009-01-01 00:40:00    996.51     -8.31    265.12        -9.07    94.2
4 2009-01-01 00:50:00    996.51     -8.27    265.15        -9.04    94.1

   VPmax (mbar)  VPact (mbar)  VPdef (mbar)  sh (g/kg)  H2OC (mmol/mol)  \
0          3.33          3.11          0.22       1.94             3.12
1          3.23          3.02          0.21       1.89             3.03
2          3.21          3.01          0.20       1.88             3.02
3          3.26          3.07          0.19       1.92             3.08
4          3.27          3.08          0.19       1.92             3.09

   rho (g/m**3)  wv (m/s)  max. wv (m/s)  wd (deg)
0       1307.75      1.03           1.75     152.3
1       1309.80      0.72           1.50     136.1
2       1310.24      0.19           0.63     171.6
3       1309.19      0.34           0.50     198.0
4       1309.00      0.32           0.63     214.3
```

[6]:
```
data.sort_values(by='Date Time', inplace=True)
```

[7]:
```
duplicates = data['Date Time'].duplicated(keep=False) # gives True or False for␣
↪all rows.
data[duplicates].head() # print the rows for the indices where duplicate is␣
↪True.
```

[7]:
```
                Date Time  p (mbar)  T (degC)  Tpot (K)  Tdew (degC)  rh (%)  \
78766 2010-07-01 00:10:00    992.06     17.87    291.69        14.06    78.4
78622 2010-07-01 00:10:00    992.06     17.87    291.69        14.06    78.4
78623 2010-07-01 00:20:00    992.02     17.82    291.65        14.03    78.5
78767 2010-07-01 00:20:00    992.02     17.82    291.65        14.03    78.5
78624 2010-07-01 00:30:00    992.04     17.92    291.75        14.09    78.3

       VPmax (mbar)  VPact (mbar)  VPdef (mbar)  sh (g/kg)  H2OC (mmol/mol)  \
78766         20.50         16.07          4.43      10.14            16.20
78622         20.50         16.07          4.43      10.14            16.20
78623         20.44         16.04          4.39      10.12            16.17
78767         20.44         16.04          4.39      10.12            16.17
78624         20.57         16.10          4.46      10.16            16.23

       rho (g/m**3)  wv (m/s)  max. wv (m/s)  wd (deg)
78766       1180.21      0.31           0.56     51.11
78622       1180.21      0.31           0.56     51.11
78623       1180.38      0.23           0.48     52.64
78767       1180.38      0.23           0.48     52.64
```

```
78624       1179.97      0.18            0.40     22.10
```

[8]:
```
data = data.drop_duplicates(subset='Date Time', keep='first')
data = data.reset_index(drop=True)
```

[9]:
```
gaps = data['Date Time'].diff() != pd.Timedelta(minutes=10)
skips = data[gaps]

print(f"Number of skips: {skips.shape[0]}")
print("Rows after skips (showing next entries after each gap):")
print(skips['Date Time'].head())
```

```
Number of skips: 7
Rows after skips (showing next entries after each gap):
0        2009-01-01 00:10:00
40378    2009-10-08 10:10:00
229875   2013-05-16 09:10:00
293229   2014-07-30 08:20:00
301346   2014-09-25 09:00:00
Name: Date Time, dtype: datetime64[ns]
```

[10]:
```
# 1. Set Date Time as index (if not already)
data = data.set_index('Date Time')

# 2. Create a complete datetime index with 10-minute frequency
full_index = pd.date_range(start=data.index.min(), end=data.index.max(),␣
 ↪freq='10min') # 10T means 10 min

# 3. Reindex the dataframe to include missing times
data = data.reindex(full_index)

# 4. Interpolate linearly
data = data.interpolate(method='linear')

# 5. (Optional) Reset index and rename it back
data = data.reset_index().rename(columns={'index': 'Date Time'})
```

[11]:
```
# Check if there are gaps
gaps = data['Date Time'].diff() != pd.Timedelta(minutes=10)
skips = data[gaps]

print(f"Number of skips: {skips.shape[0]}")
print("Rows after skips (showing next entries after each gap):")
print(skips['Date Time'].head())
```

```
Number of skips: 1
Rows after skips (showing next entries after each gap):
0   2009-01-01 00:10:00
```

```
Name: Date Time, dtype: datetime64[ns]
```

[12]:
```
data.isna().sum()
```

[12]:
```
Date Time          0
p (mbar)           0
T (degC)           0
Tpot (K)           0
Tdew (degC)        0
rh (%)             0
VPmax (mbar)       0
VPact (mbar)       0
VPdef (mbar)       0
sh (g/kg)          0
H2OC (mmol/mol)    0
rho (g/m**3)       0
wv (m/s)           0
max. wv (m/s)      0
wd (deg)           0
dtype: int64
```

[13]:
```
data = data.set_index('Date Time')
data_1H = data.resample('1h').mean()

data_1H.head()
```

[13]:
```
                      p (mbar)  T (degC)    Tpot (K)  Tdew (degC)     rh (%)  \
Date Time
2009-01-01 00:00:00  996.528000 -8.304000  265.118000    -9.120000  93.780000
2009-01-01 01:00:00  996.525000 -8.065000  265.361667    -8.861667  93.933333
2009-01-01 02:00:00  996.745000 -8.763333  264.645000    -9.610000  93.533333
2009-01-01 03:00:00  996.986667 -8.896667  264.491667    -9.786667  93.200000
2009-01-01 04:00:00  997.158333 -9.348333  264.026667   -10.345000  92.383333

                     VPmax (mbar)  VPact (mbar)  VPdef (mbar)  sh (g/kg)  \
Date Time
2009-01-01 00:00:00      3.260000      3.058000      0.202000   1.910000
2009-01-01 01:00:00      3.323333      3.121667      0.201667   1.951667
2009-01-01 02:00:00      3.145000      2.940000      0.201667   1.836667
2009-01-01 03:00:00      3.111667      2.898333      0.210000   1.811667
2009-01-01 04:00:00      3.001667      2.775000      0.231667   1.733333

                     H2OC (mmol/mol)  rho (g/m**3)  wv (m/s)  max. wv (m/s)  \
Date Time
2009-01-01 00:00:00         3.068000   1309.196000  0.520000       1.002000
2009-01-01 01:00:00         3.133333   1307.981667  0.316667       0.711667
2009-01-01 02:00:00         2.950000   1311.816667  0.248333       0.606667
2009-01-01 03:00:00         2.906667   1312.813333  0.176667       0.606667
```

```
2009-01-01 04:00:00         2.780000   1315.355000  0.290000       0.670000

                        wd (deg)
Date Time
2009-01-01 00:00:00  174.460000
2009-01-01 01:00:00  172.416667
2009-01-01 02:00:00  196.816667
2009-01-01 03:00:00  157.083333
2009-01-01 04:00:00  150.093333
```

Setting Dataloaders and Sequence generation

[14]:
```python
# Prepare data
target_data = data_1H[['T (degC)']].copy()  # don't overwrite 'data'
target_data = target_data.dropna()

# Split data (70% train, 20% val, 10% test)
n = len(target_data)
train_df = target_data[:int(n * 0.7)]
val_df = target_data[int(n * 0.7):int(n * 0.9)]
test_df = target_data[int(n * 0.9):]

# Scale data
scaler = StandardScaler()
train_scaled = scaler.fit_transform(train_df)
val_scaled = scaler.transform(val_df)
test_scaled = scaler.transform(test_df)
```

[15]:
```python
# Sequence generator
def create_sequences(data, seq_length):
    X, y = [], []
    for i in range(len(data) - seq_length):
        X.append(data[i:i+seq_length])
        y.append(data[i+seq_length])
    return np.array(X), np.array(y)

sequence_length = 24*6  # using past 6 days (24 hours * 6) to predict next hour
X_train, y_train = create_sequences(train_scaled, sequence_length)
X_val, y_val = create_sequences(val_scaled, sequence_length)
X_test, y_test = create_sequences(test_scaled, sequence_length)

# Reshape for LSTM
X_train = X_train.reshape(-1, sequence_length, 1)
X_val = X_val.reshape(-1, sequence_length, 1)
X_test = X_test.reshape(-1, sequence_length, 1)
```

[16]:
```python
print("First five temperature in sequence 1")
print(X_train[0,0:5])
```

```
print()
print("First five temperature in sequence 2")
print(X_train[1,0:5])
```

```
First five temperature in sequence 1
[[-2.01429683]
 [-1.98665073]
 [-2.06742977]
 [-2.08285298]
 [-2.13509908]]

First five temperature in sequence 2
[[-1.98665073]
 [-2.06742977]
 [-2.08285298]
 [-2.13509908]
 [-2.17770569]]
```

[17]:
```
X_train_tensor = torch.tensor(X_train, dtype=torch.float32)
y_train_tensor = torch.tensor(y_train, dtype=torch.float32)

X_val_tensor = torch.tensor(X_val, dtype=torch.float32)
y_val_tensor = torch.tensor(y_val, dtype=torch.float32)

X_test_tensor = torch.tensor(X_test, dtype=torch.float32)
y_test_tensor = torch.tensor(y_test, dtype=torch.float32)

train_dataset = TensorDataset(X_train_tensor, y_train_tensor)
val_dataset = TensorDataset(X_val_tensor, y_val_tensor)
test_dataset = TensorDataset(X_test_tensor, y_test_tensor)

batch_size = 64

train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)

val_loader = DataLoader(val_dataset, batch_size=batch_size, shuffle=False)
test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False)
```

Part B)

[18]:
```
class LSTMModel(nn.Module):
    def __init__(self):
        super(LSTMModel, self).__init__()

        self.lstm1 = nn.LSTM(input_size=1, hidden_size=64, batch_first=True)
        self.dropout1 = nn.Dropout(0.2)
```

```
        self.lstm2 = nn.LSTM(input_size=64, hidden_size=32, batch_first=True)
        self.dropout2 = nn.Dropout(0.2)

        self.dense = nn.Linear(32, 1)

    def forward(self, x):
        out, _ =  self.lstm1(x)
        out = self.dropout1(out)

        out, _ = self.lstm2(out)
        out = self.dropout2(out)

        out = self.dense(out[:, -1, :])  # take the last time step
        return out
```

[19]:
```
def train_model(model, train_loader, val_loader, optimizer, criterion,␣
↪scheduler=None, num_epochs=100):
    best_val_loss = float("inf")
    delta = 1e-5
    best_model = copy.deepcopy(model.state_dict())
    last_improvement_epoch = 0

    train_losses = []
    val_losses = []

    for epoch in range(num_epochs):
        run_epoch_train_loss = 0
        model.train()
        for X_batch, y_batch in train_loader:
            X_batch = X_batch.to(device)
            y_batch = y_batch.to(device)
            optimizer.zero_grad()
            logits = model(X_batch)
            loss = criterion(logits , y_batch)
            loss.backward()
            torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
            optimizer.step()
            run_epoch_train_loss += loss.item()

        avg_train_loss = run_epoch_train_loss / len(train_loader)
        train_losses.append(avg_train_loss)

        print(f"Epoch {epoch+1}/{num_epochs}, Train Loss: {avg_train_loss:.
↪4f}", end='')

        # Validation phase
        model.eval()
```

```
        run_epoch_val_loss = 0
        with torch.no_grad():
            for X_batch, y_batch in val_loader:
                X_batch = X_batch.to(device)
                y_batch = y_batch.to(device) # Reshape to (batch_size, 1)
                logits = model(X_batch)
                loss = criterion(logits, y_batch)
                run_epoch_val_loss += loss.item()

        avg_val_loss = run_epoch_val_loss / len(val_loader)
        val_losses.append(avg_val_loss)

        print(f", Validation Loss: {avg_val_loss:.4f}")

        if scheduler is not None:
            scheduler.step(avg_val_loss)

        # Early stopping based on validation loss
        if avg_val_loss < best_val_loss - delta:
            best_val_loss = avg_val_loss
            best_model = copy.deepcopy(model.state_dict())
            last_improvement_epoch = epoch
        elif epoch - last_improvement_epoch >= 15: # Stop if no improvement for␣
↪15 epochs
            print(f"Early stopping at epoch {epoch+1} with best validation loss:
↪ {best_val_loss:.4f}")
            break

    return train_losses, val_losses, best_val_loss, best_model,␣
↪last_improvement_epoch
```

[20]:
```
model = LSTMModel().to(device)
criterion = nn.MSELoss()
optimizer = optim.AdamW(model.parameters(), lr=0.001, weight_decay=1e-5)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='min',␣
↪factor=0.5, patience=5)
```

[21]:
```
t_l, v_l, b_v_l, b_m, l_i_e = train_model(model, train_loader, val_loader,␣
↪optimizer, criterion, scheduler, num_epochs=40)
```

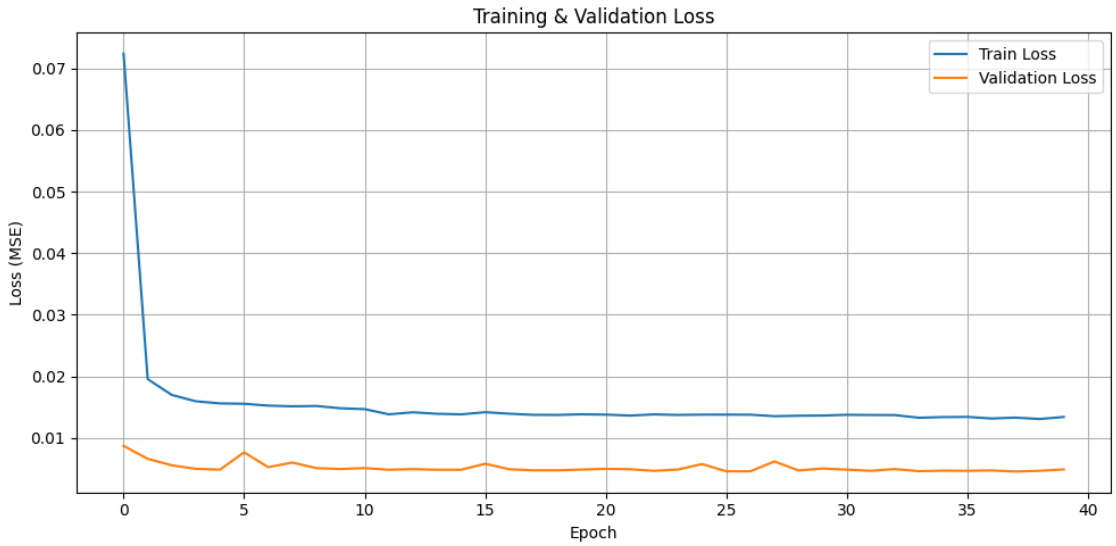
```
Epoch 1/40, Train Loss: 0.0724, Validation Loss: 0.0087
Epoch 2/40, Train Loss: 0.0196, Validation Loss: 0.0066
Epoch 3/40, Train Loss: 0.0170, Validation Loss: 0.0056
Epoch 4/40, Train Loss: 0.0160, Validation Loss: 0.0050
Epoch 5/40, Train Loss: 0.0156, Validation Loss: 0.0049
Epoch 6/40, Train Loss: 0.0156, Validation Loss: 0.0077
Epoch 7/40, Train Loss: 0.0153, Validation Loss: 0.0053
Epoch 8/40, Train Loss: 0.0152, Validation Loss: 0.0060
```

```
Epoch 9/40, Train Loss: 0.0152, Validation Loss: 0.0051
Epoch 10/40, Train Loss: 0.0148, Validation Loss: 0.0050
Epoch 11/40, Train Loss: 0.0147, Validation Loss: 0.0051
Epoch 12/40, Train Loss: 0.0139, Validation Loss: 0.0049
Epoch 13/40, Train Loss: 0.0142, Validation Loss: 0.0050
Epoch 14/40, Train Loss: 0.0139, Validation Loss: 0.0049
Epoch 15/40, Train Loss: 0.0138, Validation Loss: 0.0049
Epoch 16/40, Train Loss: 0.0142, Validation Loss: 0.0058
Epoch 17/40, Train Loss: 0.0140, Validation Loss: 0.0049
Epoch 18/40, Train Loss: 0.0138, Validation Loss: 0.0048
Epoch 19/40, Train Loss: 0.0138, Validation Loss: 0.0048
Epoch 20/40, Train Loss: 0.0139, Validation Loss: 0.0049
Epoch 21/40, Train Loss: 0.0138, Validation Loss: 0.0050
Epoch 22/40, Train Loss: 0.0136, Validation Loss: 0.0049
Epoch 23/40, Train Loss: 0.0138, Validation Loss: 0.0047
Epoch 24/40, Train Loss: 0.0138, Validation Loss: 0.0049
Epoch 25/40, Train Loss: 0.0138, Validation Loss: 0.0058
Epoch 26/40, Train Loss: 0.0138, Validation Loss: 0.0046
Epoch 27/40, Train Loss: 0.0138, Validation Loss: 0.0046
Epoch 28/40, Train Loss: 0.0135, Validation Loss: 0.0062
Epoch 29/40, Train Loss: 0.0136, Validation Loss: 0.0047
Epoch 30/40, Train Loss: 0.0137, Validation Loss: 0.0051
Epoch 31/40, Train Loss: 0.0138, Validation Loss: 0.0049
Epoch 32/40, Train Loss: 0.0137, Validation Loss: 0.0047
Epoch 33/40, Train Loss: 0.0137, Validation Loss: 0.0050
Epoch 34/40, Train Loss: 0.0133, Validation Loss: 0.0046
Epoch 35/40, Train Loss: 0.0134, Validation Loss: 0.0047
Epoch 36/40, Train Loss: 0.0134, Validation Loss: 0.0047
Epoch 37/40, Train Loss: 0.0132, Validation Loss: 0.0047
Epoch 38/40, Train Loss: 0.0133, Validation Loss: 0.0046
Epoch 39/40, Train Loss: 0.0131, Validation Loss: 0.0047
Epoch 40/40, Train Loss: 0.0134, Validation Loss: 0.0049
```

[22]:
```
model = LSTMModel().to(device)
model.load_state_dict(b_m)
```

[22]:
```
<All keys matched successfully>
```

[23]:
```
plt.figure(figsize=(10, 5))
plt.plot(t_l, label='Train Loss')
plt.plot(v_l, label='Validation Loss')
plt.xlabel("Epoch")
plt.ylabel("Loss (MSE)")
plt.title("Training & Validation Loss")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



[24]:
```
from sklearn.metrics import mean_squared_error, mean_absolute_error,␣
↪mean_absolute_percentage_error

with torch.no_grad():
    model.eval()
    y_pred_test = model(X_test_tensor.to(device)).cpu().numpy()
    y_test_np = y_test_tensor.cpu().numpy()
    y_pred_test_inv = scaler.inverse_transform(y_pred_test)
    y_test_inv = scaler.inverse_transform(y_test)
mse = mean_squared_error(y_test_inv, y_pred_test_inv)
mae = mean_absolute_error(y_test_inv, y_pred_test_inv)
mape = mean_absolute_percentage_error(y_test_inv, y_pred_test_inv)
print(f"Test MSE: {mse:.2f}, MAE: {mae:.2f},MAPE: {mape:.2%}")
```

```
Test MSE: 0.33, MAE: 0.40,MAPE: 10.73%
```

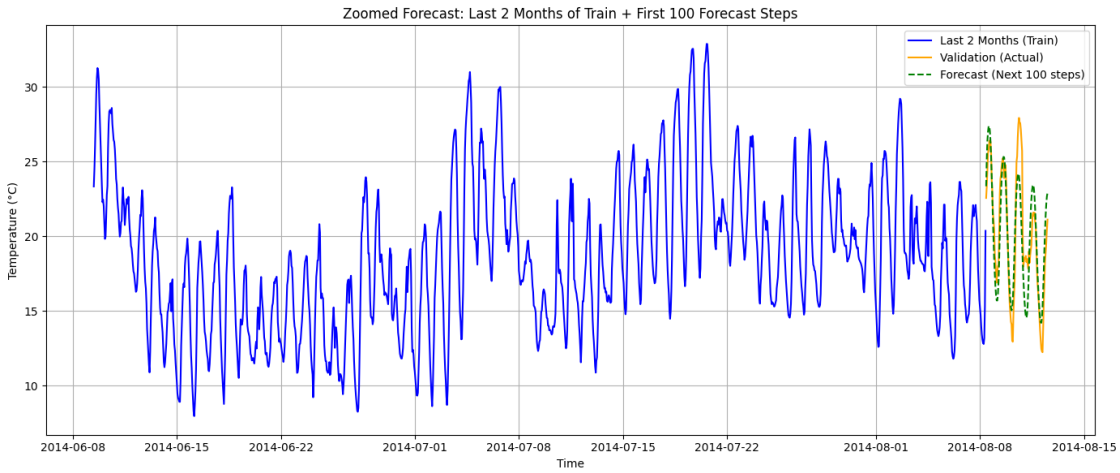
Part C)

[25]:
```
# Forecast the next 100 Values
n_forecast = 100
model.eval()
last_sequence = train_scaled[-sequence_length:].reshape(1, sequence_length, 1)
forecast_scaled = []

for _ in range(n_forecast):
    next_pred = model(torch.tensor(last_sequence, dtype=torch.float32).
↪to(device)).cpu().detach().numpy()
    forecast_scaled.append(next_pred[0, 0])
```

```
    last_sequence = np.append(last_sequence[:, 1:, :], [[[next_pred[0, 0]]]],␣
 ↪axis=1)


# Inverse scale forecast
forecast = scaler.inverse_transform(np.array(forecast_scaled).reshape(-1, 1))

# Build forecast time index
forecast_index = val_df.index[:n_forecast]
```

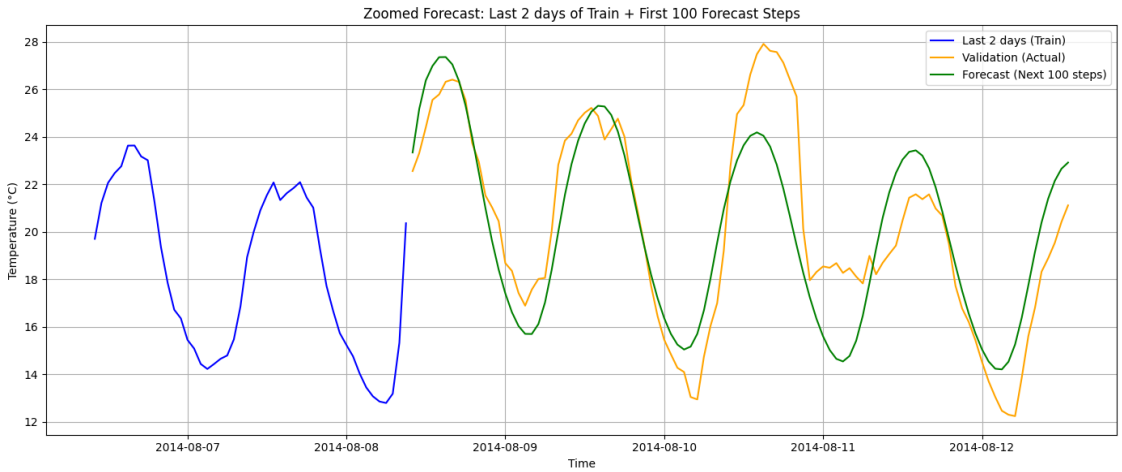
[32]:
```
# Slice last 2 months of training data
last_2_months = train_df.last('60D')  # or '2M' for full calendar months

# First 100 validation actuals
val_plot = val_df.iloc[:n_forecast]

# Plot
plt.figure(figsize=(14, 6))
plt.plot(last_2_months.index, last_2_months['T (degC)'], label='Last 2 Months␣
 ↪(Train)', color='blue')
plt.plot(val_plot.index, val_plot['T (degC)'], color='orange',␣
 ↪label='Validation (Actual)')
plt.plot(forecast_index, forecast, label='Forecast (Next 100 steps)',␣
 ↪color='green',linestyle='--')

plt.xlabel("Time")
plt.ylabel("Temperature (°C)")
plt.title("Zoomed Forecast: Last 2 Months of Train + First 100 Forecast Steps")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
C:\Users\baldi\AppData\Local\Temp\ipykernel_3096\1818456642.py:2: FutureWarning:
last is deprecated and will be removed in a future version. Please create a mask
and filter using `.loc` instead
  last_2_months = train_df.last('60D')  # or '2M' for full calendar months
```



[33]:
```
last_2_days = train_df.last('2D')

# First 100 validation actuals
val_plot = val_df.iloc[:n_forecast]

# Plot
plt.figure(figsize=(14, 6))
plt.plot(last_2_days.index, last_2_days['T (degC)'], label='Last 2 days␣
↪(Train)', color='blue')
plt.plot(val_plot.index, val_plot['T (degC)'], color='orange',␣
↪label='Validation (Actual)')
plt.plot(forecast_index, forecast, label='Forecast (Next 100 steps)',␣
↪color='green')

plt.xlabel("Time")
plt.ylabel("Temperature (°C)")
plt.title("Zoomed Forecast: Last 2 days of Train + First 100 Forecast Steps")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
C:\Users\baldi\AppData\Local\Temp\ipykernel_3096\1151203900.py:1: FutureWarning:
last is deprecated and will be removed in a future version. Please create a mask
and filter using `.loc` instead
  last_2_days = train_df.last('2D')
```



Part d)

[28]:
```
class LSTMModel2(nn.Module):
    def __init__(self):
        super(LSTMModel2, self).__init__()

        self.lstm1 = nn.LSTM(input_size=1, hidden_size=128, batch_first=True)
        self.dropout1 = nn.Dropout(0.3)


        self.lstm2 = nn.LSTM(input_size=128, hidden_size=64, batch_first=True)
        self.dropout2 = nn.Dropout(0.3)

        self.lstm3 = nn.LSTM(input_size=64, hidden_size=64, batch_first=True)
        self.dropout3 = nn.Dropout(0.3)

        self.dense = nn.Linear(64, 1)

    def forward(self, x):
        out, _ =  self.lstm1(x)
        out = self.dropout1(out)

        out, _ = self.lstm2(out)
        out = self.dropout2(out)

        out, _ = self.lstm3(out)
        out = self.dropout3(out)

        out = self.dense(out[:, -1, :])  # take the last time step
        return out
```

[29]:
```
model2 = LSTMModel2().to(device)
optimizer2 = optim.AdamW(model2.parameters(), lr=0.001, weight_decay=
1e-5)
scheduler2 = optim.lr_scheduler.ReduceLROnPlateau(optimizer2, mode='min',␣
↪factor=0.5, patience=5)
```

Model Training)

[30]:
```
t_l2, v_l2, b_v_l2, b_m2, l_i_e2 = train_model(model2, train_loader,␣
↪val_loader, optimizer2, criterion, scheduler2, num_epochs=60)
```

```
Epoch 1/60, Train Loss: 0.0676, Validation Loss: 0.0071
Epoch 2/60, Train Loss: 0.0188, Validation Loss: 0.0080
Epoch 3/60, Train Loss: 0.0170, Validation Loss: 0.0052
Epoch 4/60, Train Loss: 0.0157, Validation Loss: 0.0059
Epoch 5/60, Train Loss: 0.0157, Validation Loss: 0.0059
Epoch 6/60, Train Loss: 0.0155, Validation Loss: 0.0052
Epoch 7/60, Train Loss: 0.0156, Validation Loss: 0.0055
Epoch 8/60, Train Loss: 0.0151, Validation Loss: 0.0056
Epoch 9/60, Train Loss: 0.0148, Validation Loss: 0.0052
Epoch 10/60, Train Loss: 0.0149, Validation Loss: 0.0056
Epoch 11/60, Train Loss: 0.0151, Validation Loss: 0.0051
Epoch 12/60, Train Loss: 0.0148, Validation Loss: 0.0052
Epoch 13/60, Train Loss: 0.0146, Validation Loss: 0.0050
Epoch 14/60, Train Loss: 0.0148, Validation Loss: 0.0056
Epoch 15/60, Train Loss: 0.0146, Validation Loss: 0.0050
Epoch 16/60, Train Loss: 0.0146, Validation Loss: 0.0067
Epoch 17/60, Train Loss: 0.0145, Validation Loss: 0.0062
Epoch 18/60, Train Loss: 0.0145, Validation Loss: 0.0052
Epoch 19/60, Train Loss: 0.0143, Validation Loss: 0.0077
Epoch 20/60, Train Loss: 0.0134, Validation Loss: 0.0052
Epoch 21/60, Train Loss: 0.0132, Validation Loss: 0.0047
Epoch 22/60, Train Loss: 0.0136, Validation Loss: 0.0053
Epoch 23/60, Train Loss: 0.0135, Validation Loss: 0.0052
Epoch 24/60, Train Loss: 0.0133, Validation Loss: 0.0052
Epoch 25/60, Train Loss: 0.0136, Validation Loss: 0.0063
Epoch 26/60, Train Loss: 0.0131, Validation Loss: 0.0048
Epoch 27/60, Train Loss: 0.0132, Validation Loss: 0.0052
Epoch 28/60, Train Loss: 0.0129, Validation Loss: 0.0046
Epoch 29/60, Train Loss: 0.0127, Validation Loss: 0.0049
Epoch 30/60, Train Loss: 0.0128, Validation Loss: 0.0053
Epoch 31/60, Train Loss: 0.0129, Validation Loss: 0.0049
Epoch 32/60, Train Loss: 0.0129, Validation Loss: 0.0049
Epoch 33/60, Train Loss: 0.0127, Validation Loss: 0.0050
Epoch 34/60, Train Loss: 0.0127, Validation Loss: 0.0046
Epoch 35/60, Train Loss: 0.0128, Validation Loss: 0.0047
Epoch 36/60, Train Loss: 0.0127, Validation Loss: 0.0047
Epoch 37/60, Train Loss: 0.0126, Validation Loss: 0.0047
```

```
Epoch 38/60, Train Loss: 0.0128, Validation Loss: 0.0047
Epoch 39/60, Train Loss: 0.0128, Validation Loss: 0.0049
Epoch 40/60, Train Loss: 0.0128, Validation Loss: 0.0046
Epoch 41/60, Train Loss: 0.0124, Validation Loss: 0.0048
Epoch 42/60, Train Loss: 0.0125, Validation Loss: 0.0045
Epoch 43/60, Train Loss: 0.0126, Validation Loss: 0.0047
Epoch 44/60, Train Loss: 0.0127, Validation Loss: 0.0046
Epoch 45/60, Train Loss: 0.0126, Validation Loss: 0.0048
Epoch 46/60, Train Loss: 0.0124, Validation Loss: 0.0045
Epoch 47/60, Train Loss: 0.0124, Validation Loss: 0.0048
Epoch 48/60, Train Loss: 0.0124, Validation Loss: 0.0045
Epoch 49/60, Train Loss: 0.0124, Validation Loss: 0.0045
Epoch 50/60, Train Loss: 0.0124, Validation Loss: 0.0045
Epoch 51/60, Train Loss: 0.0126, Validation Loss: 0.0045
Epoch 52/60, Train Loss: 0.0126, Validation Loss: 0.0046
Epoch 53/60, Train Loss: 0.0124, Validation Loss: 0.0045
Epoch 54/60, Train Loss: 0.0124, Validation Loss: 0.0046
Epoch 55/60, Train Loss: 0.0122, Validation Loss: 0.0047
Epoch 56/60, Train Loss: 0.0124, Validation Loss: 0.0046
Epoch 57/60, Train Loss: 0.0122, Validation Loss: 0.0045
Epoch 58/60, Train Loss: 0.0121, Validation Loss: 0.0046
Epoch 59/60, Train Loss: 0.0123, Validation Loss: 0.0046
Epoch 60/60, Train Loss: 0.0122, Validation Loss: 0.0045
```

[31]:
```
model2 = LSTMModel2().to(device)
model2.load_state_dict(b_m2)
```

[31]:
```
<All keys matched successfully>
```

[34]:
```
plt.figure(figsize=(10, 5))
plt.plot(t_l2, label='Train Loss')
plt.plot(v_l2, label='Validation Loss')
plt.xlabel("Epoch")
plt.ylabel("Loss (MSE)")
plt.title("Training & Validation Loss")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

Metrics

[37]:
```
with torch.no_grad():
    model2.eval()
    batch_size = 256
    preds = []
    for i in range(0, len(X_test_tensor), batch_size):
        batch= X_test_tensor[i:i+batch_size].to(device)
        preds.append(model2(batch).cpu().numpy())
    y_pred_test = np.concatenate(preds, axis=0)
    y_test_np = y_test_tensor.cpu().numpy()
    y_pred_test_inv = scaler.inverse_transform(y_pred_test)
    y_test_inv = scaler.inverse_transform(y_test)
mse = mean_squared_error(y_test_inv, y_pred_test_inv)
mae = mean_absolute_error(y_test_inv, y_pred_test_inv)
mape = mean_absolute_percentage_error(y_test_inv, y_pred_test_inv)
print(f"Test MSE: {mse:.2f}, MAE: {mae:.2f},MAPE: {mape:.2%}")
```

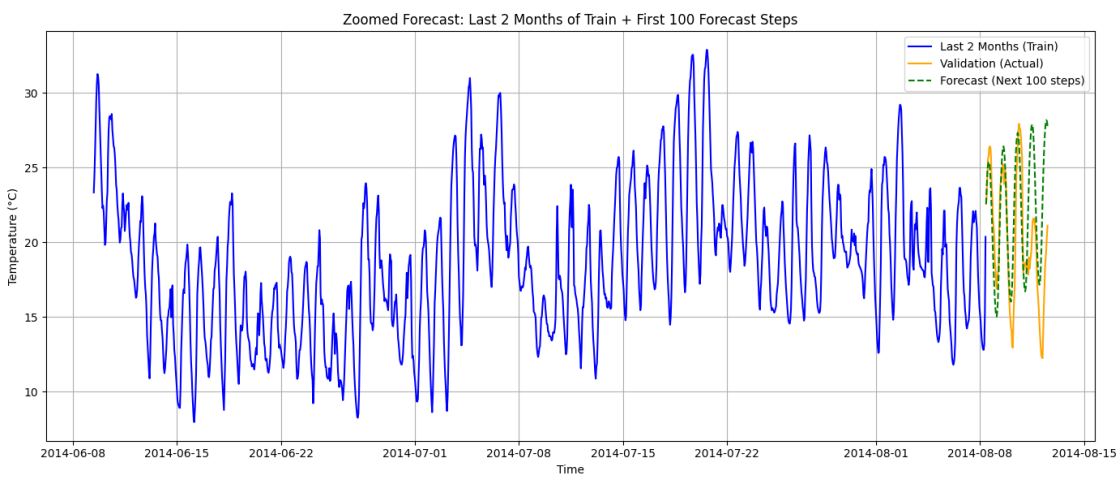
```
Test MSE: 0.33, MAE: 0.40,MAPE: 10.47%
```

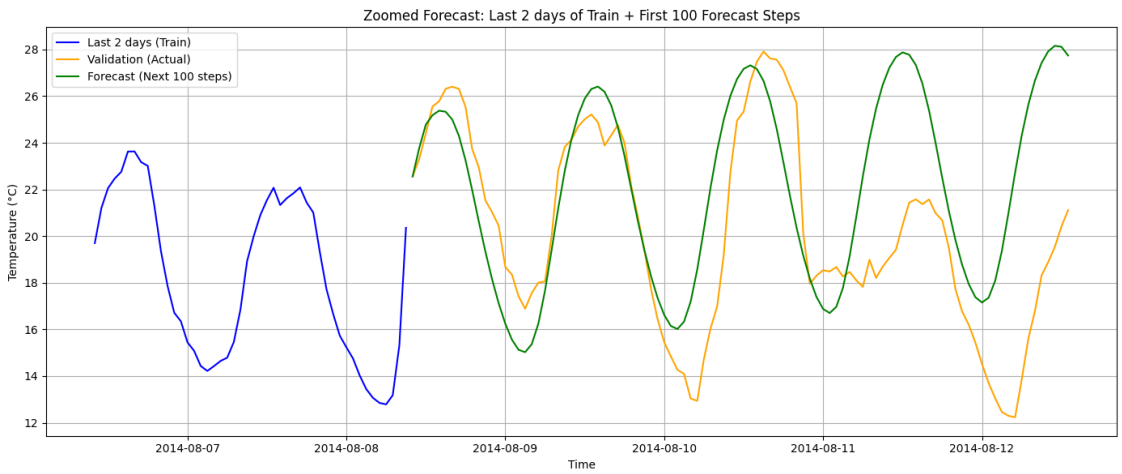
Forecasting)

[38]:
```
model2.eval()
last_sequence_2 = train_scaled[-sequence_length:].reshape(1, sequence_length, 1)
forecast_scaled_2 = []

for _ in range(n_forecast):
    next_pred = model2(torch.tensor(last_sequence_2, dtype=torch.float32).
 ↪to(device)).cpu().detach().numpy()
    forecast_scaled_2.append(next_pred[0, 0])
```

```
    last_sequence_2 = np.append(last_sequence_2[:, 1:, :], [[[next_pred[0,␣
↪0]]]], axis=1)


# Inverse scale forecast
forecast_2 = scaler.inverse_transform(np.array(forecast_scaled_2).reshape(-1,␣
↪1))

# Build forecast time index
forecast_index_2 = val_df.index[:n_forecast]
```

[39]:
```
# Plot
plt.figure(figsize=(14, 6))
plt.plot(last_2_months.index, last_2_months['T (degC)'], label='Last 2 Months␣
↪(Train)', color='blue')
plt.plot(val_plot.index, val_plot['T (degC)'], color='orange',␣
↪label='Validation (Actual)')
plt.plot(forecast_index_2, forecast_2, label='Forecast (Next 100 steps)',␣
↪color='green',linestyle='--')

plt.xlabel("Time")
plt.ylabel("Temperature (°C)")
plt.title("Zoomed Forecast: Last 2 Months of Train + First 100 Forecast Steps")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



[40]:
```
# Plot
plt.figure(figsize=(14, 6))
```

```
plt.plot(last_2_days.index, last_2_days['T (degC)'], label='Last 2 days␣
↪(Train)', color='blue')
plt.plot(val_plot.index, val_plot['T (degC)'], color='orange',␣
↪label='Validation (Actual)')
plt.plot(forecast_index_2, forecast_2, label='Forecast (Next 100 steps)',␣
↪color='green')

plt.xlabel("Time")
plt.ylabel("Temperature (°C)")
plt.title("Zoomed Forecast: Last 2 days of Train + First 100 Forecast Steps")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```